

Internship Mid Report

Weather Alerts System

Shapefiles and QGIS:

- The pakistan.geojson file contains the exact latitude and longitude coordinate arrays that define the country's borders. Used react's mapping capabilities to parse it into an SVG layer for UI purposes.
- Made use of Ray casting for point in polygon assurance both for this and later for aircraft mapping to make sure no aircraft data is being used that is outside the Pakistani border.

Data Formats:

- Hourly data pulled from OpenMeteo API, timestamped and contains variables like *temperature_2m, relative_humidity_2m, precipitation, and wind_speed_10m*.
- Normalisation of this data by doing things like extracting timezones and scaling was all done via pandas DataFrame so that training the AI model later doesn't cause issues.
- Also made a custom windows scheduler via batch file that automatically pulls latest 24hrs weather data and updates the CSV logs and updates weights as part of automation.

AI prediction model:

- Standard neural networks treat all inputs independently, which fails for time-series forecasting so I used LSTM as it learns which past data to remember and which to discard, to spot long-term sequential patterns.
- It predicts across four multivariate features (Temperature, Wind, Humidity, Rain). The model was trained over 10 years of historical data for 5 epochs using GPU acceleration to ensure rapid compilation.
- I used the "darts" library over ARIMA or scikit-learn as it allowed me to train a single deep learning model simultaneously across all five provincial capitals using its grouped dataframe capabilities. This means the model learns from the interactions of the entire country, not just isolated cities.
- Performed Cloud Training, saved the model weights to a .pt and .ckpt file, downloaded those files, and then set up Local Inference via FastAPI server.